

League of Legends Champion Strength Analysis

Jianan Hong
SMGT 530

Background

- Background:
 - Platform champion ranking tier algorithm is a black box
 - Special counter
- Research Question:
 - Can a Bradley–Terry model recover a explainable tier ranking for top-lane champions?
 - Any stats method can help to identify special counter?

Champion tier calculation explained



Bushy

4 months ago · Updated

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Q. How is champion tier calculated?

Champion tier is calculated using a proprietary algorithm that analyzes ranked game data from Platinum tier and above. A variety of factors are included in the calculation, such as pick-ban rate, win rate, gold, experience, crowd control, KDA, damage taken, and damage dealt.

Filters can be applied for server, tier, and patch version to view specific statistical data.

Global

Emerald +

Version: 14.06

All

Top

Jungle

Middle

Bottom

Support

Search a Champion

*

Aatrox

Ahri

Akali

Akshan

Alistar

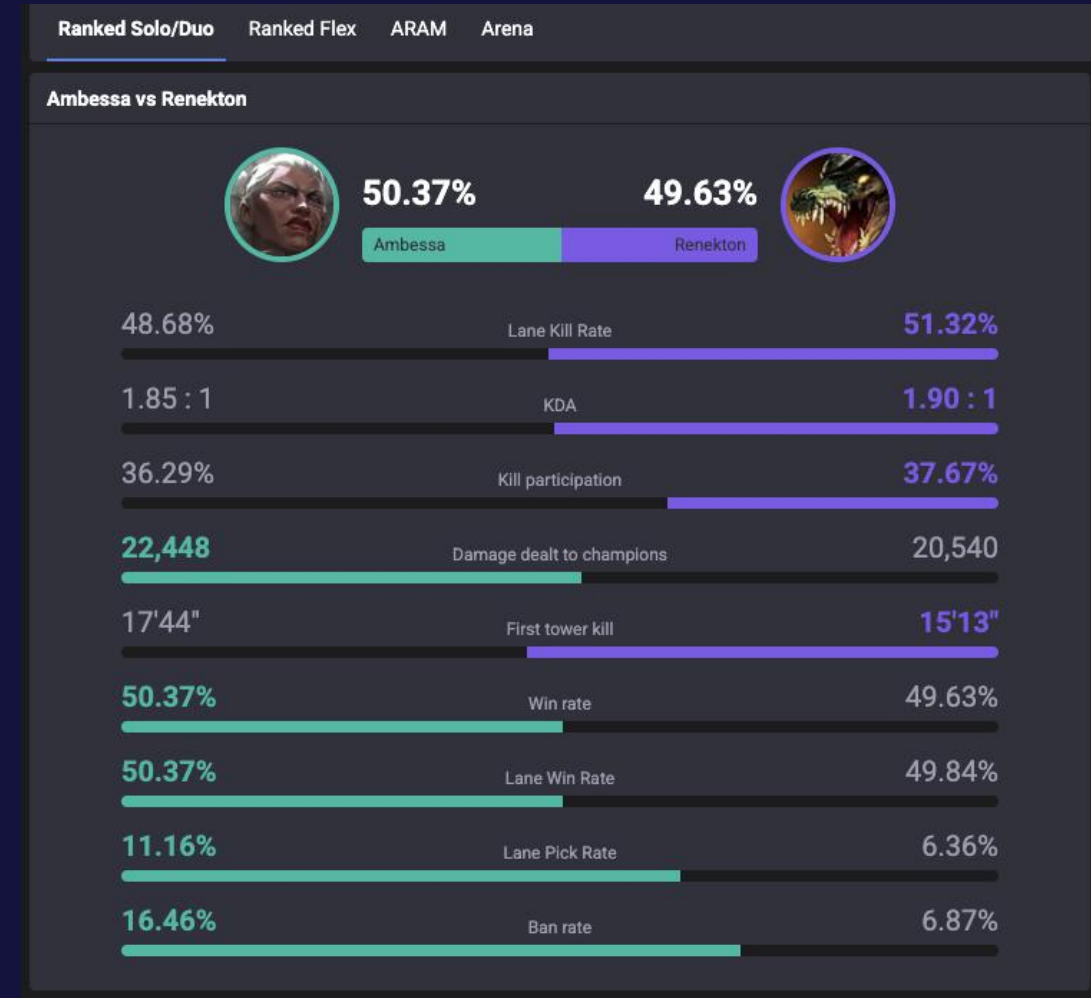
Amumu

Rank	Champion	Tier	Position	Win Rate	Pick Rate	Ban Rate	Weak Against
1	Jinx	OP		52.57%	23.11%	5.26%	
2	Ahri	OP		51.78%	17.24%	13.98%	
3	Cassiopeia	I		54.63%	0.94%	0.55%	
4	Nilah	I		54.41%	2.84%	3.84%	
5	Kayle	I		53.51%	2.78%	0.98%	

Data Source

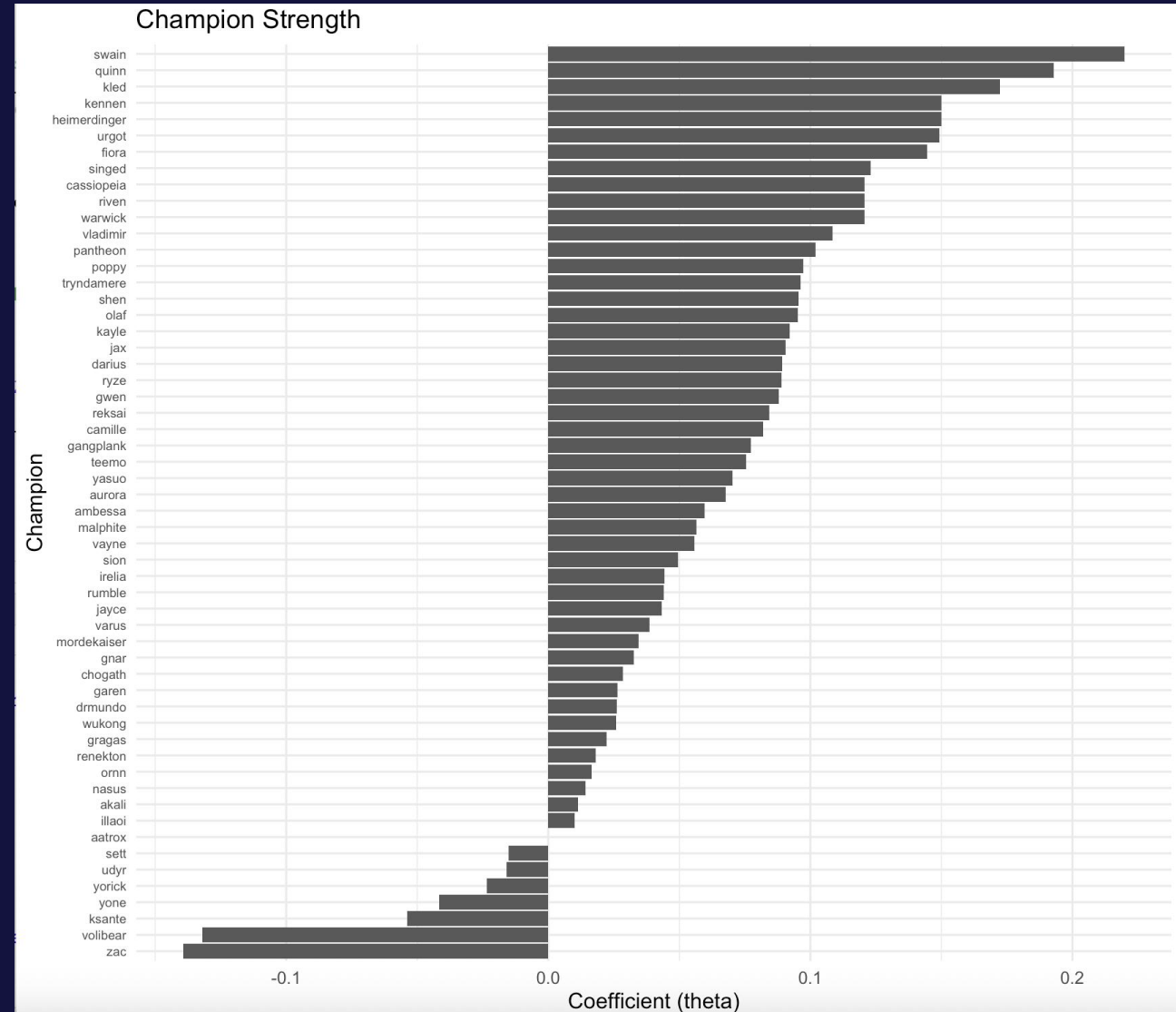
- Source: op.gg
- Lane: Top
- ELO: Master+
- Tool: Python scraping
- Sample data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	champion	opponent	games	win_rate	lane_kill_rate	kda	kill_particip	damage_to	first_tower	lane_win_r	lane_pick	ban_rate	url	
2	ambessa	atrox	719	51.95%	51.24%	1.95 : 1	37.72%	24,339	16'34"	50.54%	10.03%	14.73%	https://op.gg/lol/	
3	ambessa	jax	519	48.46%	53.85%	2.05 : 1	38.27%	22,496	16'12"	50.54%	10.03%	14.73%	https://op.gg/lol/	
4	ambessa	sion	583	51.89%	55.47%	2.22 : 1	38.13%	28,217	17'43"	50.54%	10.03%	14.73%	https://op.gg/lol/	
5	ambessa	ksante	678	55.33%	45.02%	1.94 : 1	35.87%	23,207	17'37"	50.54%	10.03%	14.73%	https://op.gg/lol/	
6	ambessa	renekton	1387	51.22%	47.59%	1.88 : 1	36.66%	22,296	17'33"	50.54%	10.03%	14.73%	https://op.gg/lol/	
7	ambessa	jayce	500	45.69%	51.01%	1.75 : 1	36.91%	18,043	16'33"	50.54%	10.03%	14.73%	https://op.gg/lol/	
8	ambessa	fiora	363	47.91%	54.11%	1.91 : 1	36.97%	23,281	16'15"	50.54%	10.03%	14.73%	https://op.gg/lol/	
9	ambessa	irelia	466	46.90%	52.39%	2.00 : 1	39.15%	21,187	17'18"	50.54%	10.03%	14.73%	https://op.gg/lol/	
10	ambessa	malphite	317	50.16%	49.41%	2.11 : 1	37.35%	20,775	17'0"	50.54%	10.03%	14.73%	https://op.gg/lol/	
11	ambessa	gwen	396	47.10%	57.01%	2.10 : 1	39.36%	22,826	16'30"	50.54%	10.03%	14.73%	https://op.gg/lol/	



Strength –Results I

- Raw BT model
 - Raw dataset is highly biased
 - Benefit small individuals
- How to improve?

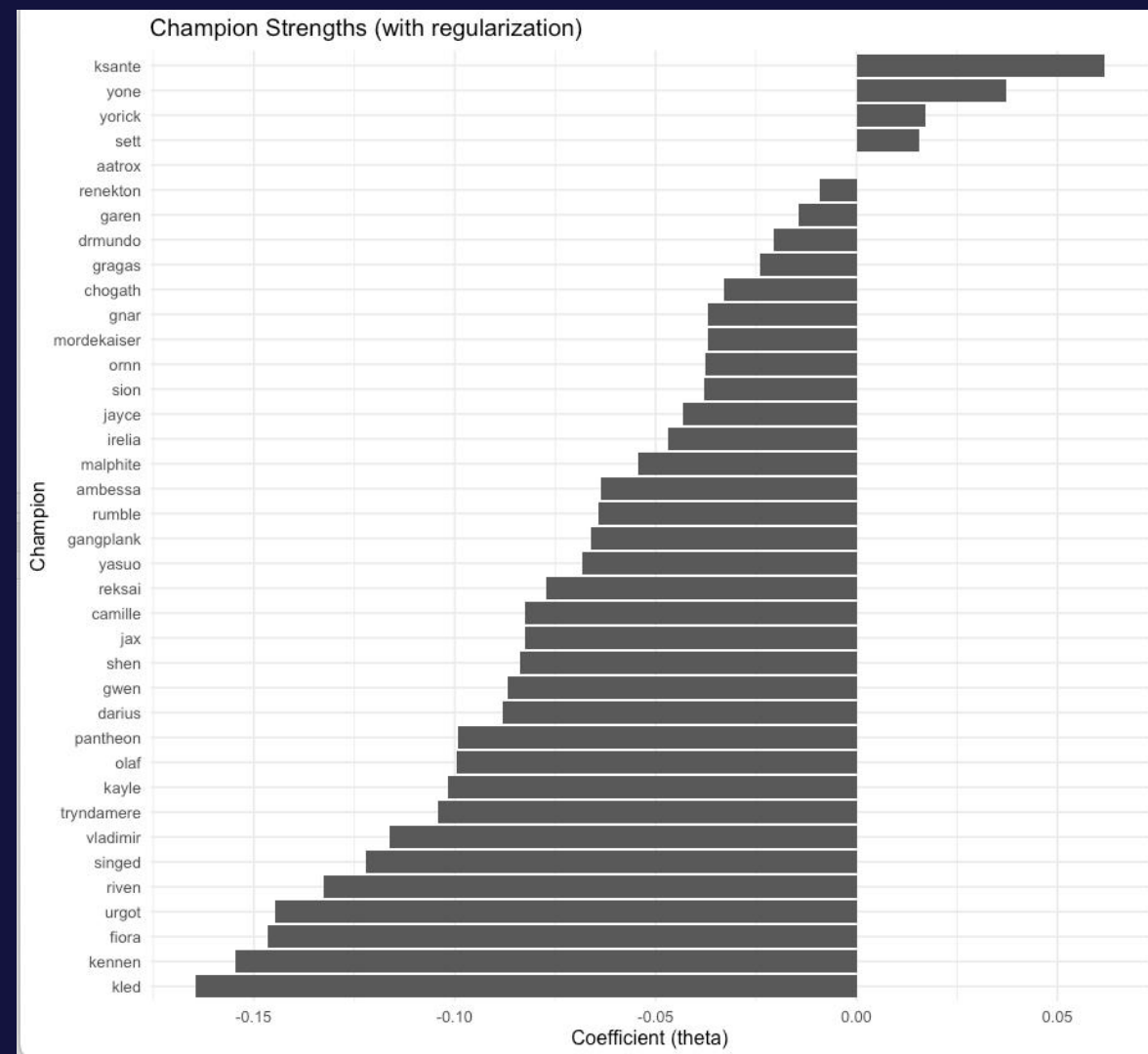


Strength –Results II

- Regularized BT model

$$\hat{\beta} = \arg \min_{\beta} - \sum_{i=1}^n \left(z_i \log \left(\frac{e^{\eta_i}}{1 + e^{\eta_i}} \right) + (1 - z_i) \log \left(\frac{1}{1 + e^{\eta_i}} \right) \right) + \lambda \sum_{j=2}^p \beta_j^2$$

- First filtered low pick rate champions (threshold 2%)
 - Lambda: cv_ridge\$lambda.min (lowest mean cross-validation error)
 - Works better!
- Example:
 - K'sante vs. Ornn
 - Win rate made by BT model: 52.2%
 - Real win rate: 53.5%
 - Compare with OPGG
 - Top 5 from BTm: K'sante, Yone, Yorick, Sett, Aatrox
 - Top 5 from OPGG: Ambessa, Aatrox, K'sante, Jax, Sion



Counters –Results I

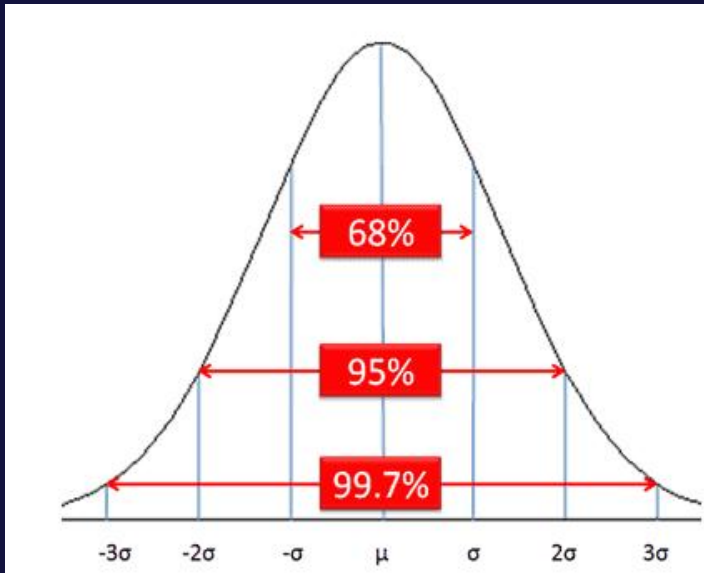
- Special Counters

- Hypothesis:

- H0: $\hat{p}_{c,o} = p_c$

- Ha: $\hat{p}_{c,o} \neq p_c$

- Using Z score; threshold $z = 2$ (95%)



$$p_c = \frac{\text{total wins of champion } c}{\text{total games of champion } c}$$

$$\hat{p}_{c,o} = \frac{\text{wins of champion } c \text{ vs opponent } o}{n_{c,o}}$$

$$z_{c,o} = \frac{\hat{p}_{c,o} - p_c}{SE_{c,o}}$$

- Large Z score $\rightarrow$ overperform
- Low Z score $\rightarrow$ underperform

Counters –Result II

	champion	opponent	n_games	p_obs	p_exp	z_score	p_value	lane_pick_rate
1	mordekaiser	volibear	79	0.6962	0.4961655	3.555995	0.0003765505	0.0422
2	fiora	ksante	384	0.6084	0.5227494	3.360284	0.0007786243	0.0516
3	yone	ksante	234	0.5823	0.4757318	3.264206	0.0010977115	0.0401
4	sion	yorick	139	0.6312	0.4981378	3.137582	0.0017034756	0.0764
5	jax	darius	290	0.5972	0.5084958	3.021593	0.0025144826	0.0686
6	ayale	ksante	221	0.6091	0.5106455	2.927927	0.0034123040	0.0246
7	cassiopeia	mordekaiser	37	0.7568	0.5192555	2.891999	0.0038279862	0.0150
8	akali	yone	84	0.6429	0.4880425	2.839397	0.0045198933	0.0194
9	atrox	warwick	98	0.6263	0.4855478	2.787917	0.0053048212	0.0918
10	ambessa	ksante	678	0.5533	0.5015003	2.697580	0.0069845582	0.1003
11	gragas	gnar	76	0.6447	0.4908434	2.683032	0.0072958032	0.0313
12	yorick	jayce	173	0.5799	0.4784260	2.671854	0.0075433548	0.0232
13	shen	ksante	246	0.5935	0.5098296	2.625145	0.0086612174	0.0394
14	warwick	mordekaiser	35	0.7353	0.5161101	2.594837	0.0094635865	0.0151
15	singed	drundo	91	0.6522	0.5165926	2.588649	0.0096353117	0.0291
16	illaoi	ornd	58	0.6607	0.4922179	2.566554	0.0102714691	0.0118
17	ksante	nasus	70	0.6232	0.4709393	2.552123	0.0107068587	0.0808
18	swain	malphite	31	0.7742	0.5465394	2.546175	0.0108910585	0.0082
19	gangplank	singed	117	0.6228	0.5053611	2.540738	0.0110618921	0.0365
20	jax	mordekaiser	187	0.6011	0.5084958	2.533053	0.0113073834	0.0686

Special Counter for All Champions

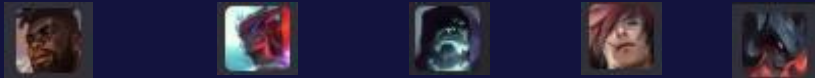
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1	cassiopeia	mordekaiser	37	0.7568	0.5192555	2.891999	0.003827986	0.0150
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4	illaoi	ornd	58	0.6607	0.4922179	2.566554	0.010271469	0.0118
5	swain	malphite	31	0.7742	0.5465394	2.546175	0.010891059	0.0082
6	vayne	camille	51	0.6731	0.5002926	2.468184	0.013580041	0.0157
7	aurora	yone	33	0.7188	0.5046174	2.460875	0.013859845	0.0148
8	poppy	ornd	62	0.6667	0.5120603	2.435977	0.014851623	0.0170
9	aurora	ksante	112	0.6195	0.5046174	2.431710	0.015027746	0.0148
10	teemo	ambessa	127	0.6032	0.5057597	2.196338	0.028067779	0.0162
11	aurora	darius	46	0.6596	0.5046174	2.102376	0.035520376	0.0148
12	nasus	yasuo	56	0.6250	0.4873354	2.061037	0.039299517	0.0172

Special Counter for low pick rate Champions

Discussion & Further Improvement

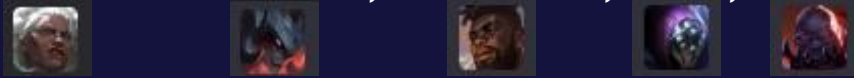
- Top 5 from BTm:

- K'sante, Yone, Yorick, Sett, Aatrox



- Top 5 from OPGG:

- Ambessa Aatrox, K'sante, Jax, Sion



- Why?

- BT model doesn't consider the games, only focus on win/loss

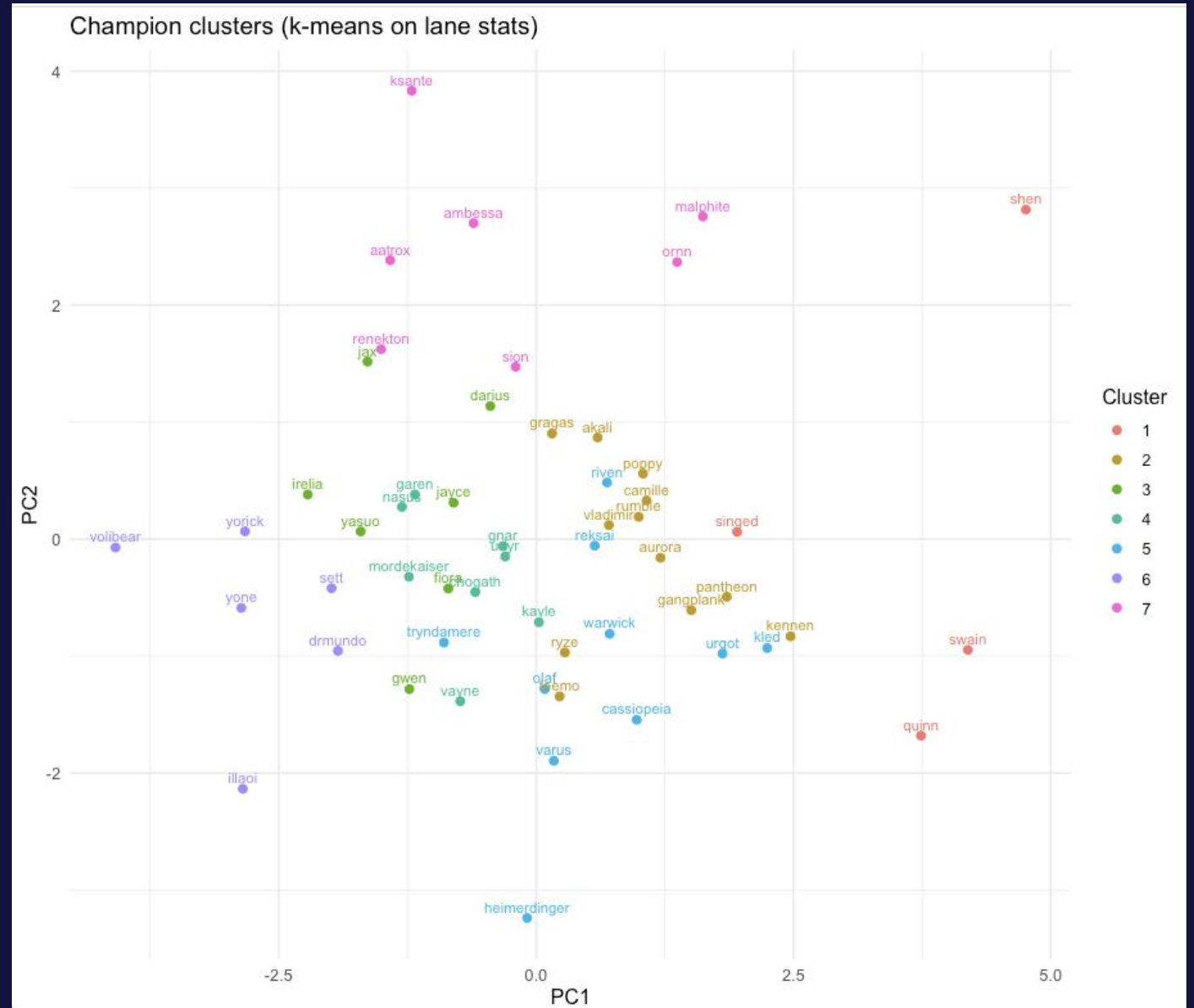
- Further Improvement

- Adding weight by games



Still have time?

- K-means cluster (for fun)
 - Make similar champions in 1 cluster
 - Using all data (10-d)
 - X,Y axis doesn't mean anything
 - Hard to explain



Questions?